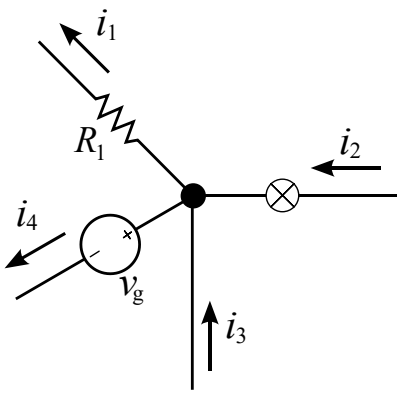
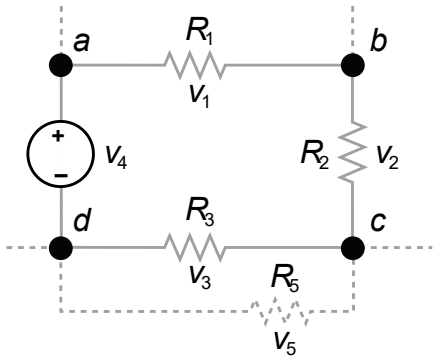
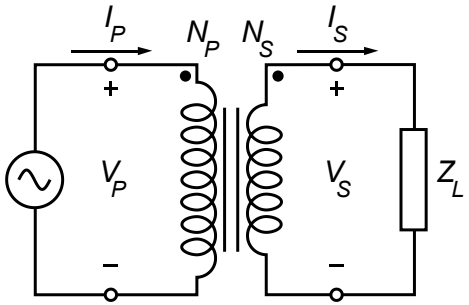
Hochschule Kaiserslautern
University of Applied Sciences

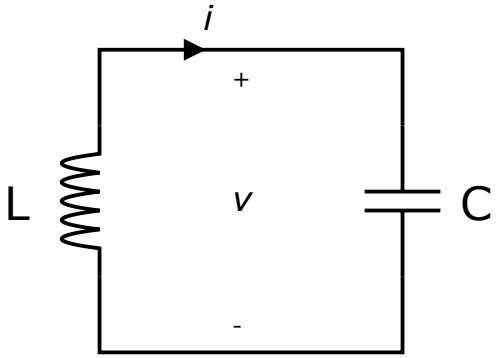
Introduction to Electrical Engineering







$$\frac{V_P}{V_S} = \frac{I_S}{I_P} = \frac{N_P}{N_S}$$



Overview

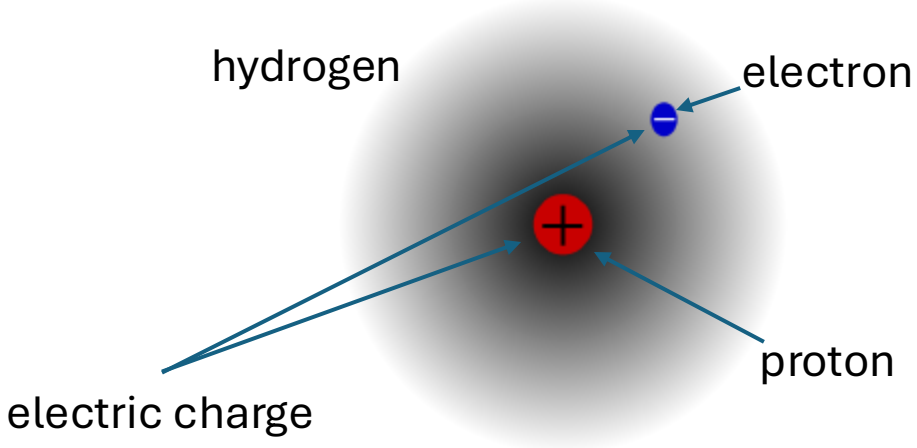
1. **Physical Principles and Basic Concepts**
2. Direct Current Circuits
3. Electric and Magnetic Fields
4. Alternating Current Technology
5. Wrap-Up and Outlook

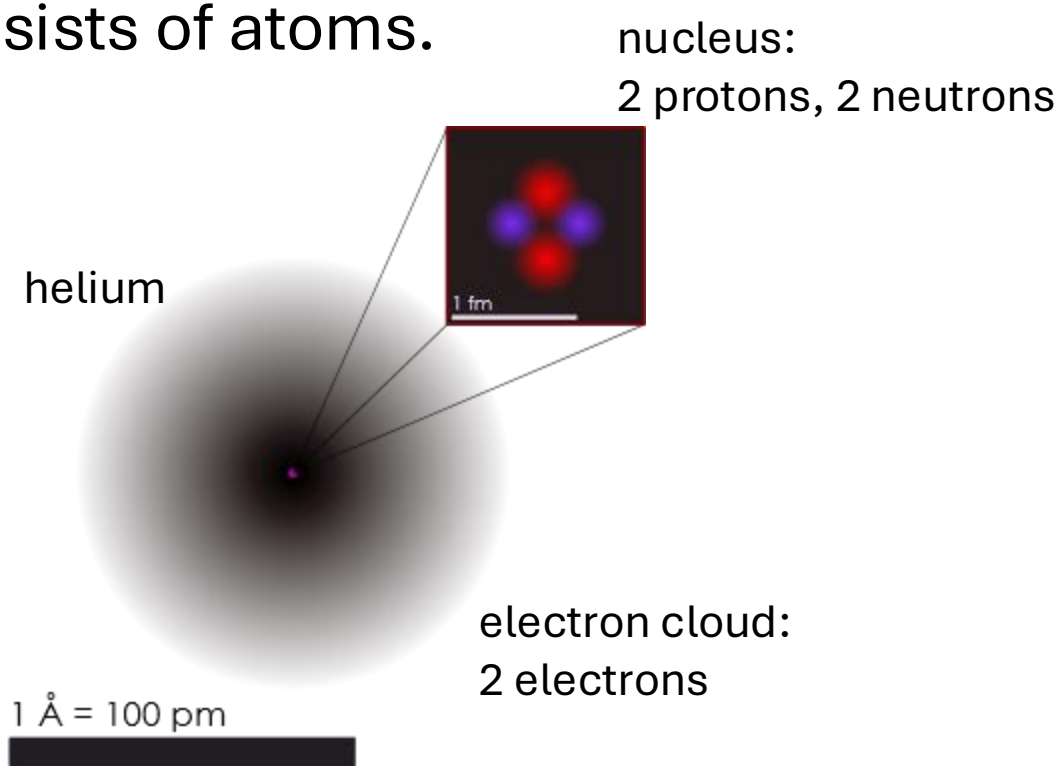
Physical Principles and Basic Concepts

- Electric Charge
- Band Structure
- Current
- Voltage
- Ohm‘s Law
- Resistors
- Ideal Voltage & Current Sources
- Electric Power

The World Is Not What It Seems

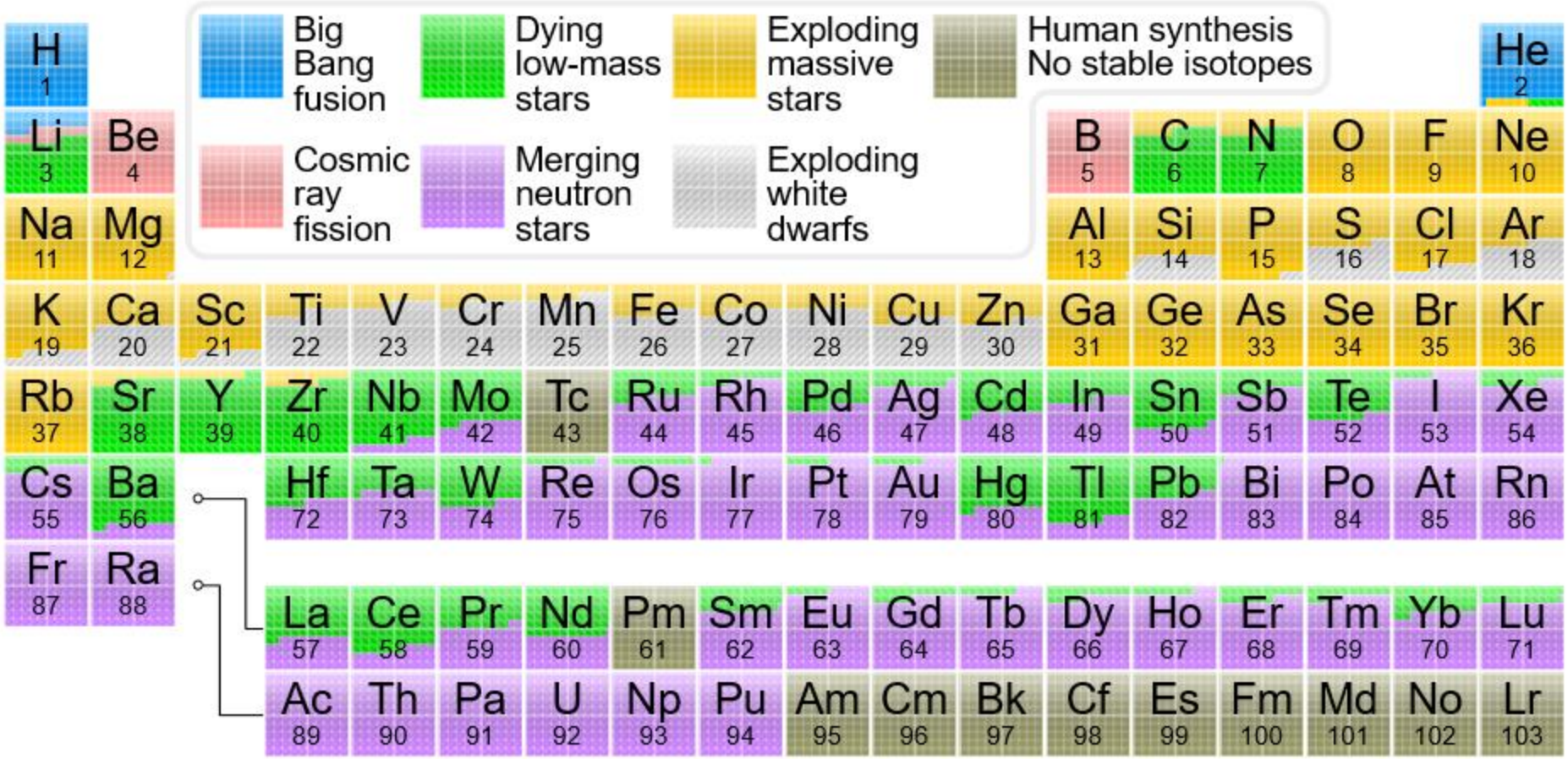
All the “stuff“ around us consists of atoms.





And atoms are nearly empty.

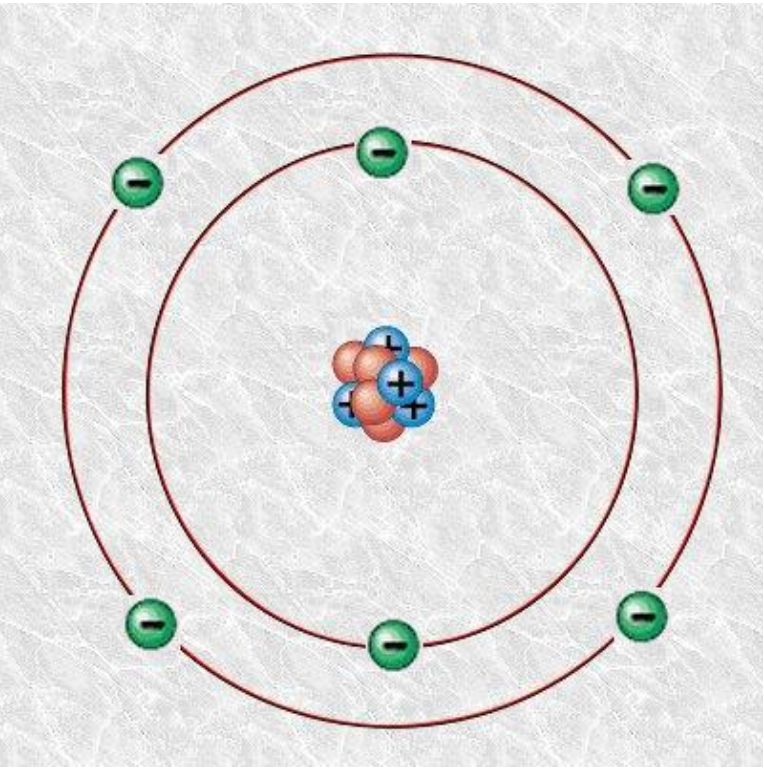
Periodic Table of Elements



Coulomb Force

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{r^2}$$

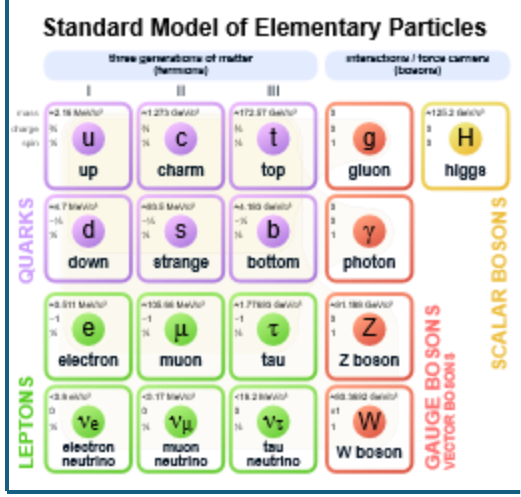
$$\varepsilon_0 = 8{,}854 \cdot 10^{-12}\,\frac{\mathrm{As}}{\mathrm{Vm}}$$



positive and negative charges attract each other

simple atomic model: balance with centrifugal force
→ stable electron orbits

Electric Charge



$$Q = n \cdot e$$

elemantary charge: $e = 1{,}6021892 \cdot 10^{-19}\ \mathrm{C}$

$$[Q] = 1\ \mathrm{C} \qquad \text{(coulomb)}$$

e: fundamental physical property

(odd number because some physical units were arbitrarily determined)

other fundamental forces: gravity (mass, only one not quantized?), strong (keeps atomic nuclei together) and weak (e.g., reason for radioactive beta decay) nuclear forces

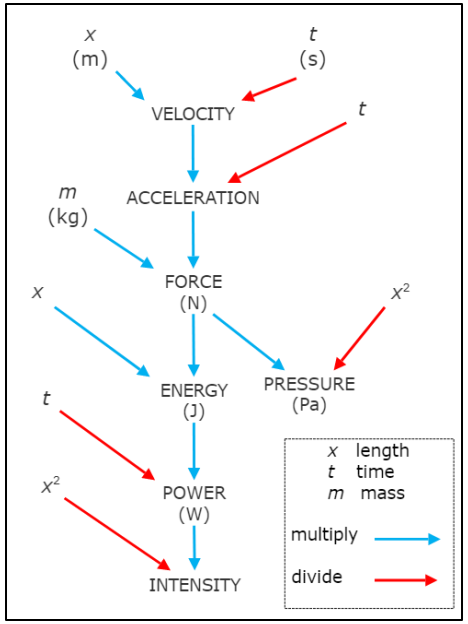
not relevant for electrical engineering though: gravity very weak, nuclear forces very short-ranged

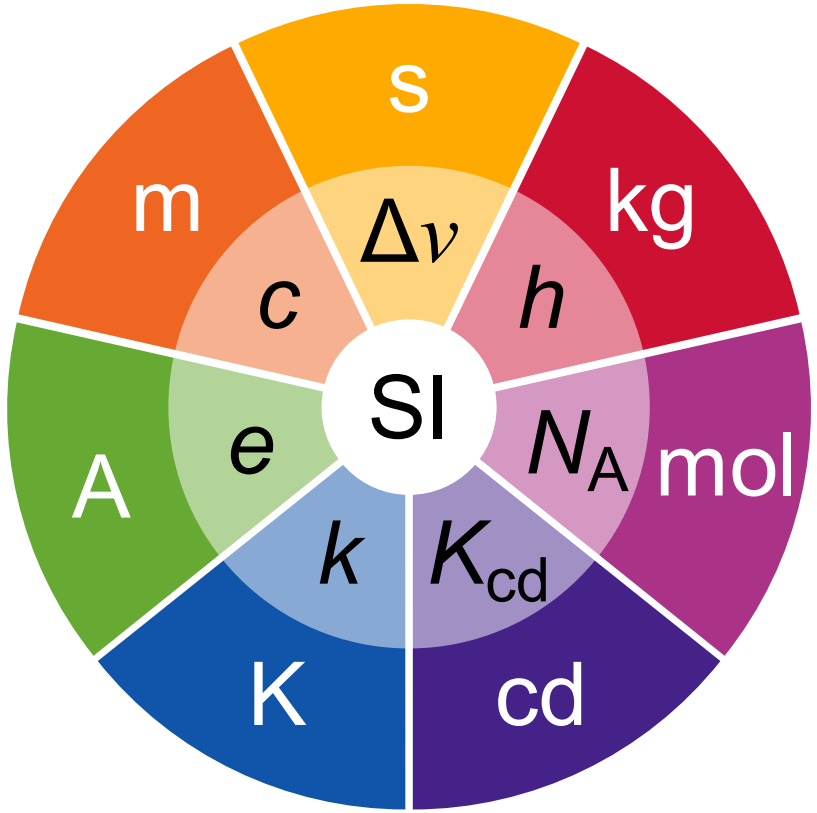
International Systems of Units (SI)

just a convention

- time
- length
- mass
- electric current
- thermodynamic temperature
- amount of substance
- luminous intensity

all other units derived

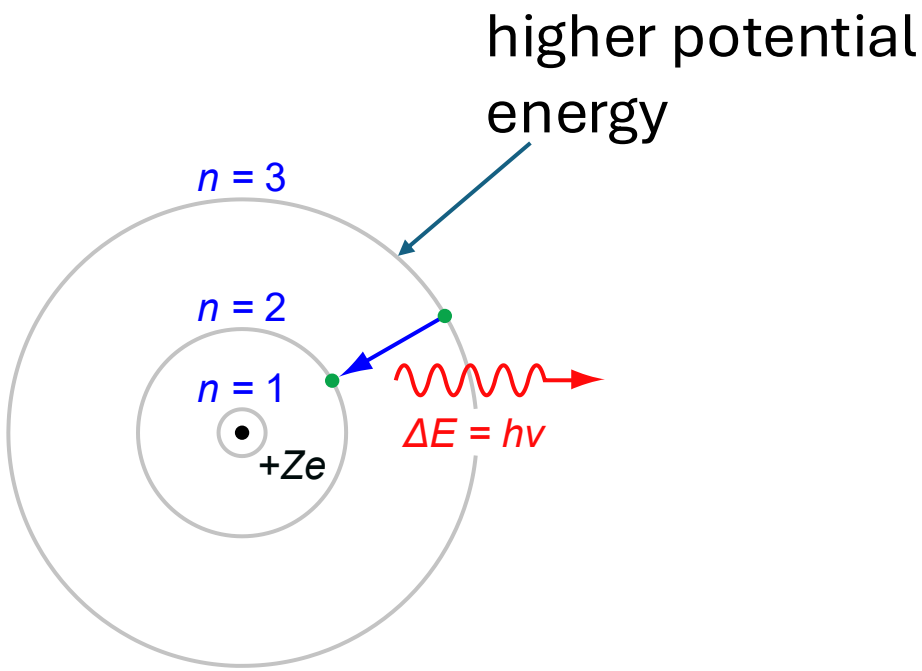




outer ring: SI base units
inner ring: constants

Bohr Model

allowing only certain energies (orbits) of electrons → quantized!



modern view: orbital model (location of electrons in atom determined by quantum-mechanical wave function → probability distribution)

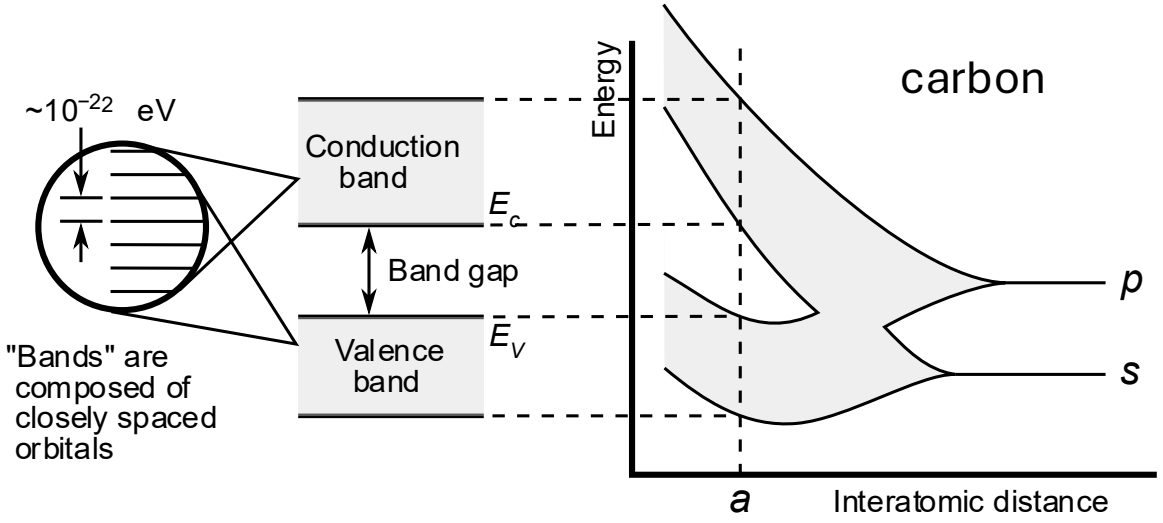
models just try to approximate the world: more complex models capture more details

Band Structure

Atomic nuclei have fixed positions in solids.

But the electron shells overlap.

→ Higher-energetic electrons can move (drift) through solid matter.



Electric Current

moving electrons → moving charges: electric current

$$I = \frac{Q}{t}$$

$$i(t) = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

$$[I] = 1\ \mathrm{A} \quad \text{(ampere)}$$

Drift Velocity

typically, $1.5\ \mathrm{mm}^2$ copper cable used for lighting and $2.5\ \mathrm{mm}^2$ for power circuits

$$Q = e \cdot n \cdot A \cdot l$$

$$\rightarrow v_{\mathrm{drift}} = \frac{I}{enA}$$

typically, $16\ \mathrm{A}$ fuses in households

example:

copper $\left(n = 8{,}49 \cdot 10^{19}\ {}^{1}/_{\mathrm{mm}^3}\right)$ wire of $A = 1.5\ \mathrm{mm}^2$ with $I = 15\ \mathrm{A}$

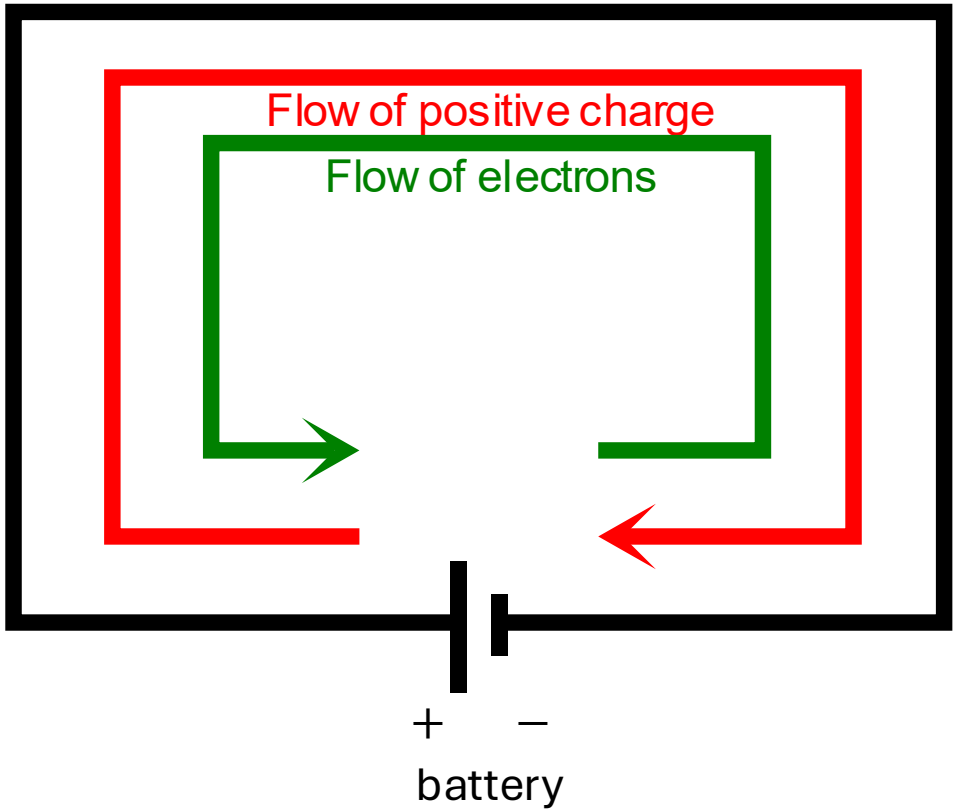
$\rightarrow v_{\mathrm{drift}} = 0.74\,\frac{\mathrm{mm}}{\mathrm{s}}$

but starts everywhere in the conductor at the same time

$J = \frac{I}{A}$ is current density (measure of thermal stress on conductor)

Conventional Current

arbitrarily defined as direction in which positive charges flow



Voltage

Voltage is the driving force of electric current.

relative quantity: potential difference between two points in space

$$V = V_{12} = \varphi(P_1) - \varphi(P_2)$$

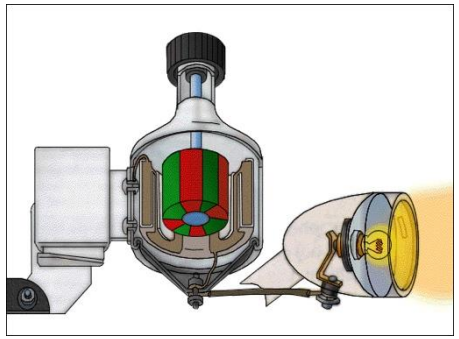
symbol U in German literature

electric potential $\varphi = \frac{W}{Q}$: defined from potential energy of a charge

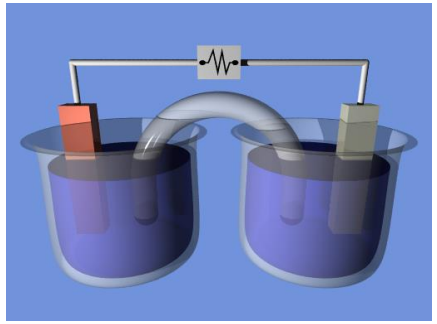
$$[V] = [\varphi] = 1\ \mathrm{V} \quad \text{(volt)}$$

Types of Voltage Generation

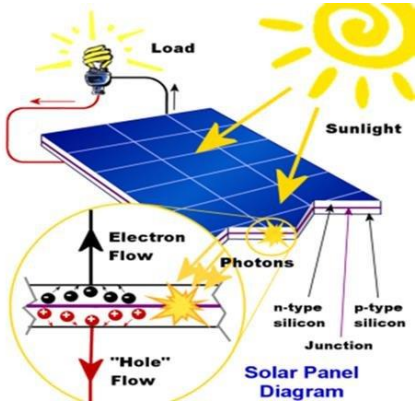
- generators (mechanical, often via turbines, e.g., from thermal energy)



- batteries (chemical)

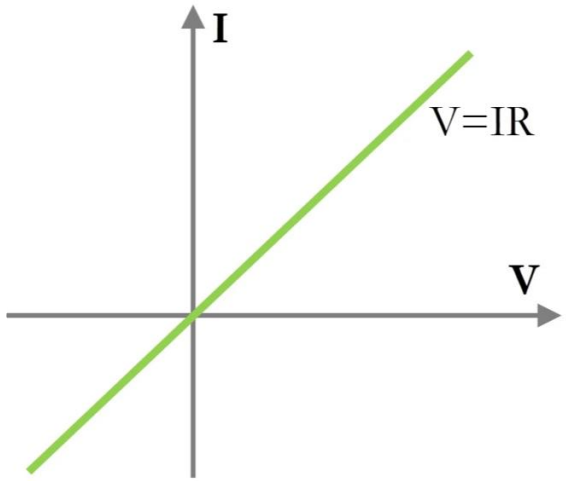


- photovoltaic



- piezoelectric
- …

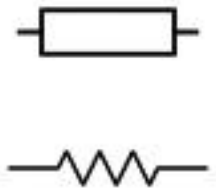
Ohm‘s Law



current-voltage characteristic
(I-V curve)

$$V = R \cdot I$$

$$[R] = 1\ \Omega \qquad \text{(ohm)}$$



electrical resistance

Electrical Conductance

$$G = \frac{1}{R} = \frac{I}{V}$$

$[G] = 1\ \mathrm{S}$ (siemens)

formerly: $[G] = 1\ \mho$ (mho)

Specific Resistance

$$R = \varrho \frac{l}{A}$$

$$[\varrho] = 1\ \frac{\Omega \mathrm{mm}^2}{\mathrm{m}}$$

also called electrical resistivity

at 20 °C

material	**ϱ (Ωmm²/m)**
silver	0.016
copper (power cable)	0.017 … 0.018
gold	0.022
aluminium	0.027
constantan (Cu + Ni)	0.5
quartz glass	$7.5 \cdot 10^{23}$

example:
copper cable with $A = 2.5\ \mathrm{mm}^2$ and $l = 10\ \mathrm{m}$ has $R = 0.07\ \Omega$
→ wiring usually negligible

Specific Conductance

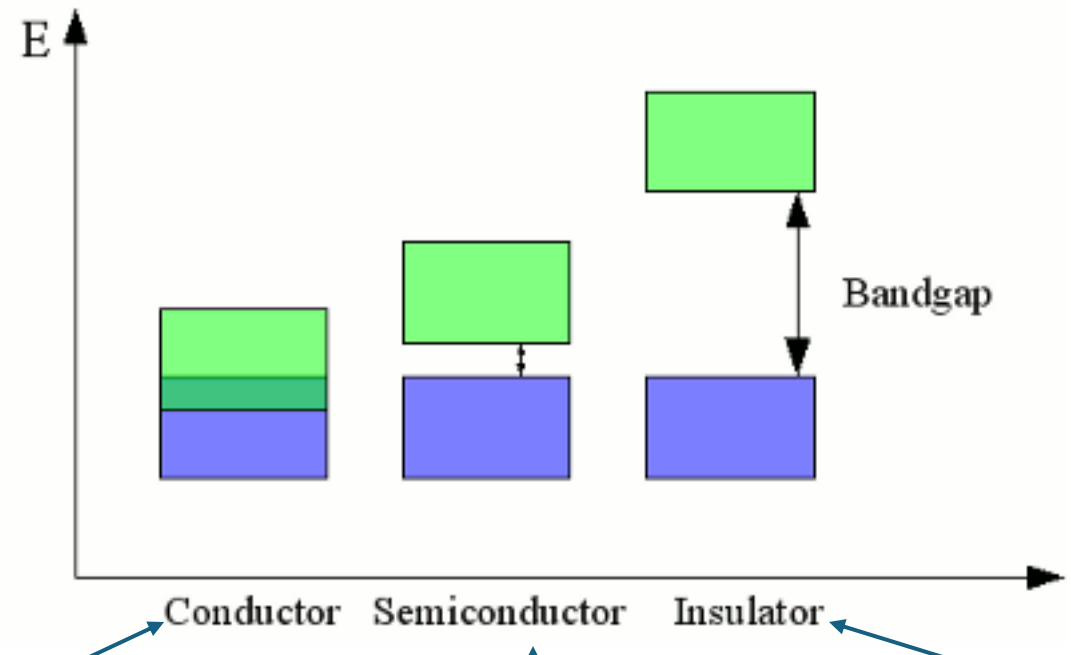
$$\kappa = \frac{1}{\varrho}$$

$$[\kappa] = 1\ \frac{\mathrm{Sm}}{\mathrm{mm}^2}$$

$$G = \kappa \frac{A}{l}$$

also called electrical conductivity

Conductor, Semiconductor, Insulator



$\kappa > 0.1 \ \frac{\mathrm{Sm}}{\mathrm{mm}^2}$

examples: copper, silver, aluminium

examples: silicon, germanium, selenium

$\varrho > 1 \cdot 10^{12} \ \frac{\Omega \mathrm{mm}^2}{\mathrm{m}}$

examples: ceramic, rubber, oil, glass

Temperature Dependence of Resistivity

- metallic conductors: resistivity increases with higher temperature (stronger vibrations of atoms, impeding the flow of electrons)

- semiconductors: resistivity decreases with higher temperature (more electrons can jump to the conduction band, increasing the number of free charge carriers, i.e., electrons and holes)

- insulators: decreases slightly with higher temperature

Temperature Coefficient

$$\varrho(T) = \varrho(T_0)\left(1 + \alpha_{T_0}(T - T_0)\right)$$

$$[\alpha] = \frac{1}{\mathrm{K}}$$

NB: linear approximation only valid in limited temperature range

for $T_0 = 20\ °\mathrm{C}$

material	α_{20} in K^{-1}
iron	0.0065
aluminium	0.0043
gold	0.0034
copper	0.0039
silver	0.0038
constantan	0.00003
germanium	-0.05
silicon	-0.07

metals: positive temperature coefficient (PTC)

semiconductors: negative temperature coefficient (NTC)

Example: Tungsten Filament

A light bulb contains a tungsten filament ($\kappa_{20} = 18.2\ \mathrm{Sm/mm^2}$, $\alpha_{20} = 4.1 \cdot 10^{-3}\ \mathrm{K^{-1}}$, $\beta_{20} = 1 \cdot 10^{-6}\ \mathrm{K^{-2}}$) with a diameter of $24\ \mu\mathrm{m}$ and a length of $62\ \mathrm{cm}$, which is operated at a constant voltage of $230\ \mathrm{V}$.

Current through the tungsten wire when the bulb is switched on in cold state ($20\ °\mathrm{C}$):

$$I_{20} = \frac{V}{R_{20}} = \frac{V}{\frac{l}{\kappa_{20} A}} = 3.05\ \mathrm{A}$$

at operating temperature ($2200\ °\mathrm{C}$):

$$I_{2200} = \frac{V}{R_{2200}} = \frac{V}{R_{20}[1 + \alpha_{20}(2200 - 20) + \underbrace{\beta_{20}(2200 - 20)^2}_{\text{quadratic term: more accurate approximation for large temperature differences}}]} = 0.21\ \mathrm{A}$$

Electrical Loads

any component/device consuming electrical power

resistive load:

electrical energy is converted into heat or light (e.g., incandescent bulbs)

resistor (also a resistive load itself):

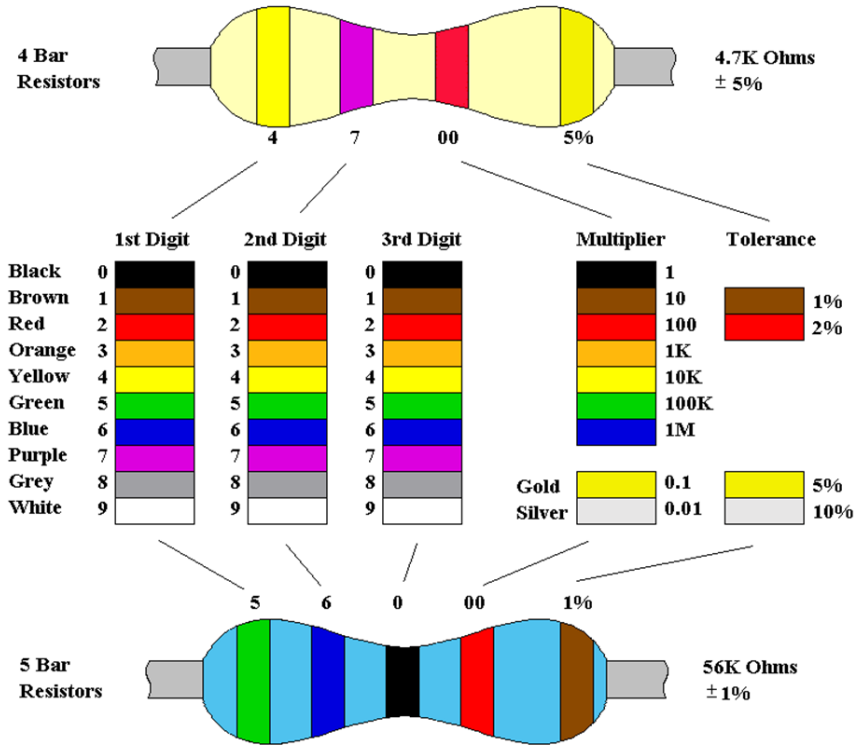
specific component to limit/control the current in electrical circuits (precise, known amount of resistance)

capacitive & inductive loads: energy temporarily stored in electric or magnetic fields

Resistors

E6 (±20%)	E12 (±10%)	E24 (±5%)
1,0	1,0	1,0
		1,1
	1,2	1,2
		1,3
1,5	1,5	1,5
		1,6
	1,8	1,8
		2,0
2,2	2,2	2,2
		2,4
	2,7	2,7
		3,0

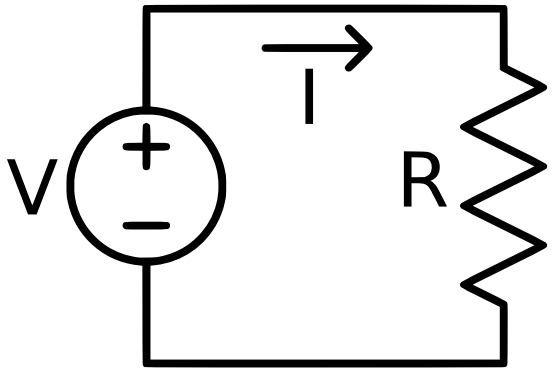
E6 (±20%)	E12 (±10%)	E24 (±5%)
3,3	3,3	3,3
		3,6
	3,9	3,9
		4,3
4,7	4,7	4,7
		5,1
	5,6	5,6
		6,2
6,8	6,8	6,8
		7,5
	8,2	8,2
		9,1



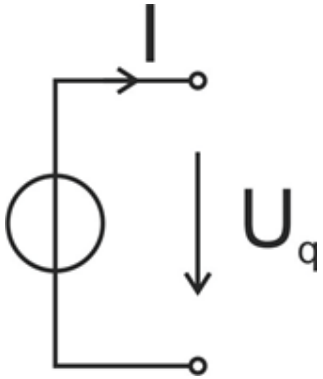
Ideal Voltage Source

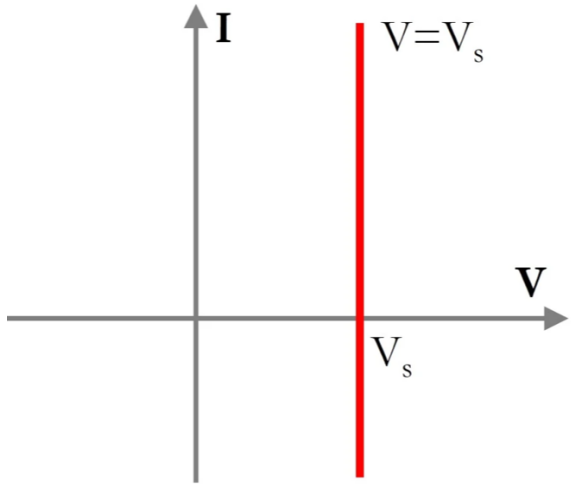
model!

maintains a fixed voltage across its two terminals (independent of the output current or load resistance)

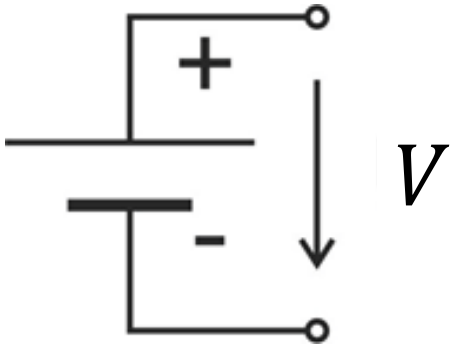


also:





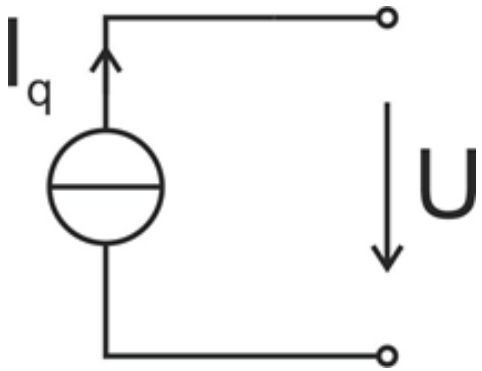
battery



Ideal Current Source

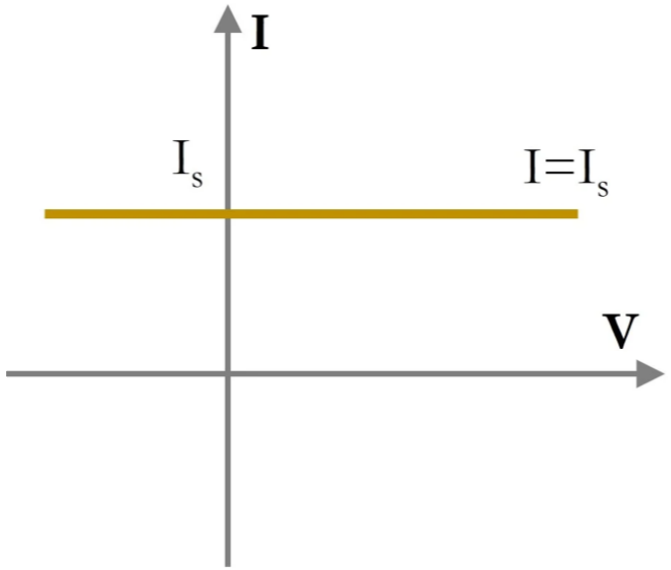
generates a fixed current (independent of the voltage changes across it)

also:



Usually, there are only voltage supplies.

However, the behavior of the current source model can be constructed using circuitry.



Work

electrical energy dissipated in a resistor:

$$W = V \cdot Q = V \cdot I \cdot t$$

if current or voltage varies over time:

$$W = \int_0^t V \cdot I \, dt$$

Electric Power

$$P = \frac{W}{t} = V \cdot I$$

$$[P] = 1\ \mathrm{W} \quad \text{(watt)}$$

example:

A cooler bag for connection to the $12\ \mathrm{V}$ electrical system of a car draws a constant current of $4\ \mathrm{A}$, resulting in $P = 48\ \mathrm{W}$. So, in 8 hours $W = 0.384\ \mathrm{kWh}$.

Summary

A basic **electrical circuit** consists of

- a **voltage source** (providing the electrical potential difference that drives the current, like a battery or generator),
- **conductive pathways** (wires or traces allowing the current to flow from the source to the load and back),
- and a **load** (device using the electrical energy, such as a light bulb, motor, or computer).

Ohm‘s law: $$I = \frac{V}{R}$$